Replication Report: (title not recorded)

OSTI 1864334 | Coverage 7/10, Agreement 7/10

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1 Source paper

- **Title:** (title not recorded)
- **Authors:** Not recorded
- **Year:** n/a
- **Domain:** Not recorded
- **Identifier:** OSTI 1864334

2 Replication objective

The central claim of the paper concerns not recorded. Our objective was to reproduce the headline numerical/qualitative results within the constraints of the AI-driven Replication Project (24–72 h, commodity GPU/CPU compute, no author contact). A replication plan is archived at `1864334-Variational-Monte-Carlo-calculations-of-A/replication_plan.pdf`.

3 Methodology

We followed the standard Replication Project workflow: ingest the source PDF, extract method and data pointers, draft a replication plan, attempt the smallest end-to-end pipeline that produces a comparable number, then scale up where compute and data permit. Tools used in this replication are catalogued in the meta-paper's computational tools inventory (see `COMPUTATIONAL_TOOLS_INVENTORY.md`).

This paper went through the Tier-Lift pipeline; supplementary notes at `.openclaw/workspace/24h-progress/tie`

Paper-directory README excerpt

Variational Monte Carlo calculations of A 4 nuclei with an artificial neural-network correlator ansatz

• **OSTI ID:** 1864334

• **Rank:** #23

• **Replication Score:** 9/10

• **Open-Source Tools:** Yes

• **Code Repository:** No

- **Tools:** PyTorch, Metropolis Monte Carlo, stochastic-reconfiguration algorithm

 # Why This Paper

Fully specified VMC/ML methodology, synthetic data, open-source tools (PyTorch), and quantitative results. No code repo but straightforward for AI.

 # Replication Plan

AI implements Hamiltonian/ansatz, uses PyTorch, generates synthetic data, optimizes network, and reproduces convergence/energy results.

 # Status

- [ ] Paper reviewed
- [ ] Code/tools identified
- [ ] Code implemented/cloned
- [ ] Results reproduced
- [ ] Results validated against paper

The replication was driven by an OpenClaw agent loop (Claude Opus / GPT-5 class models) issuing shell, Python, and (where applicable) GPU jobs. Compute usage and wall-time for individual papers are summarised in the meta-paper's resource table; not separately recorded for this paper unless noted in the Tier-Lift artefact. Sub-agents were spawned for replication-plan generation and (for papers in batches 1–4) for tier-lift extension runs.

4 What was reproduced

- Not recorded.

5 What was missing or incomplete

- Not recorded.

The dominant blocker categories for this paper, mapped onto the meta-paper's 9-category friction taxonomy (§4), are listed in §?? below.

6 Quantitative agreement

Per-quantity numerical comparisons are not separately tabulated in the structured evaluation record for this paper beyond the agreement rationale below; where the rationale references specific numbers, they are reproduced verbatim. A full numerical comparison table is recorded in the per-replication artefacts (see §??). For papers below 8/8, the agreement rationale identifies the specific quantities that diverge.

Agreement rationale. Not recorded.

7 Score and friction-point tagging

- **Coverage:** 7/10 — Not recorded.
- **Agreement:** 7/10 — Not recorded.

Friction points encountered (taxonomy tags):

- **F5**: Force-field / hyperparameter drift (unspecified configuration)

8 Follow-on questions

- Not recorded.

9 Reproducibility pointers

- **Paper directory:** `1864334-Variational-Monte-Carlo-calculations-of-A`
- **Replication artefacts:**
  - `/Users/stevens/Dropbox/REPLICATE-PROJECT/1864334-Variational-Monte-Carlo-calculations-`
- **Replication-dir hint from JSONL:** `/Users/stevens/Dropbox/REPLICATE-PROJECT/1864334-Variatio`
- **Status:** tier_lift_v2_2026_04_27

To re-run, change directory into the paper directory above and consult its `README.md` for the entry-point script. Most replications follow the convention `replication/run_*.sh` or `replication/<subdir>/run.py`.